Practical 4: Configure IP ACLs to Mitigate Attacks

R1 (Main Configuration + ACL 120)

```
enable
configure terminal
enable secret enpa55
line console 0
password conpa55
login
exit
ip domain-name ccnasecurity.com
username admin secret adminpa55
line vty 0 4
login local
transport input ssh
access-class 10 in
exit
crypto key generate rsa
1024
access-list 10 permit host 192.168.3.3
access-list 120 permit udp any host 192.168.1.3 eq domain
access-list 120 permit tcp any host 192.168.1.3 eq smtp
access-list 120 permit tcp any host 192.168.1.3 eq ftp
access-list 120 deny tcp any host 192.168.1.3 eq 443
access-list 120 permit tcp host 192.168.3.3 host 10.1.1.1 eq 22
access-list 120 permit icmp any any echo-reply
access-list 120 permit icmp any any unreachable
access-list 120 deny icmp any any
access-list 120 permit ip any any
```

```
interface se0/1/0
ip access-group 120 in
end
```

🖥️ **R2 (Loopback + OSPF + Security)**

```
enable
configure terminal
enable secret enpa55
line console 0
password conpa55
login
exit
ip domain-name ccnasecurity.com
username admin secret adminpa55
line vty 0 4
login local
transport input ssh
access-class 10 in
exit
crypto key generate rsa
1024
interface loopback 0
ip address 192.168.2.1 255.255.255.0
no shutdown
router ospf 1
network 10.1.1.0 0.0.0.3 area 0
network 10.2.2.0 0.0.0.3 area 0
access-list 10 permit host 192.168.3.3
```

end

🖥️ **R3 (ACL 110 + ACL 100)**

```
enable
configure terminal
enable secret enpa55
line console 0
password conpa55
login
exit
ip domain-name ccnasecurity.com
username admin secret adminpa55
line vty 0 4
login local
transport input ssh
access-class 10 in
exit
crypto key generate rsa
1024
router ospf 1
network 192.168.3.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.3 area 0
access-list 10 permit host 192.168.3.3
access-list 110 permit ip 192.168.3.0 0.0.0.255 any
interface gig0/1
ip access-group 110 in
access-list 100 permit tcp 10.0.0.0 0.255.255.255 host 192.168.3.3 eq 22
access-list 100 deny ip 10.0.0.0 0.255.255.255 any
```

```
access-list 100 deny ip 172.16.0.0 0.15.255.255 any
access-list 100 deny ip 192.168.0.0 0.0.255.255 any
access-list 100 deny ip 127.0.0.0 0.255.255.255 any
access-list 100 deny ip 224.0.0.0 15.255.255.255 any
access-list 100 permit ip any any
interface se0/1/0
ip access-group 100 in
end
```

✅ **Verification Commands**

```
show access-lists
show ip interface
```

Practical 5: Configure IPv6 ACLs

R1 (IPv6 + ACL BLOCK_HTTP)

```
enable
configure terminal
enable secret enpa55
ipv6 unicast-routing
interface gig0/0
ipv6 address 2001:DB8:1:10::1/64
ipv6 address FE80::1 link-local
no shutdown
interface gig0/1
ipv6 address 2001:DB8:1:11::1/64
ipv6 address FE80::1 link-local
no shutdown
interface se0/1/0
```

```
ipv6 address 2001:DB8:1:1::1/64
ipv6 address FE80::1 link-local
no shutdown
ipv6 route 2001:DB8:1:2::/64 2001:DB8:1:1::2
ipv6 route 2001:DB8:1:30::/64 2001:DB8:1:1::2
ipv6 access-list BLOCK_HTTP
deny tcp any host 2001:DB8:1:30::30 eq www
deny tcp any host 2001:DB8:1:30::30 eq 443
permit ipv6 any any
exit
interface gig0/1
ipv6 traffic-filter BLOCK_HTTP in
end
```

🖥️ **R2 (IPv6 Routing)**

```
enable
configure terminal
enable secret enpa55
ipv6 unicast-routing

interface se0/1/0
ipv6 address 2001:DB8:1:1::2/64
ipv6 address FE80::2 link-local
no shutdown
interface se0/1/1
ipv6 address 2001:DB8:1:2::2/64
ipv6 address FE80::2 link-local
no shutdown
```

```
ipv6 route 2001:DB8:1:10::/64 2001:DB8:1:1::1
ipv6 route 2001:DB8:1:11::/64 2001:DB8:1:1::1
ipv6 route 2001:DB8:1:30::/64 2001:DB8:1:2::1
end
```

🖥️ R3 (IPv6 + ACL BLOCK_ICMP)

```
enable
configure terminal
enable secret enpa55
ipv6 unicast-routing
interface gig0/0
ipv6 address 2001:DB8:1:30::1/64
ipv6 address FE80::3 link-local
no shutdown
interface se0/1/0
ipv6 address 2001:DB8:1:2::1/64
ipv6 address FE80::3 link-local
no shutdown
ipv6 route 2001:DB8:1:10::/64 2001:DB8:1:2::2
ipv6 route 2001:DB8:1:11::/64 2001:DB8:1:2::2
ipv6 route 2001:DB8:1:1::/64 2001:DB8:1:2::2
ipv6 access-list BLOCK_ICMP
deny icmp any any
permit ipv6 any any
exit
interface gig0/0
ipv6 traffic-filter BLOCK_ICMP out
end
```

✅ Verification Commands

show ipv6 access-list

show ipv6 interface